the mitotic cell cycle

25/11/24

DNA in prokaryotes	DNA in eukaryotes
circular DNA with no free ends	linear DNA with two ends
relatively short	much longer
not bound to histones	tightly wrapped around histone proteins

* DNA is also found in mitochondria & chloroplasts and it's very similar to DNA in prokaryotes

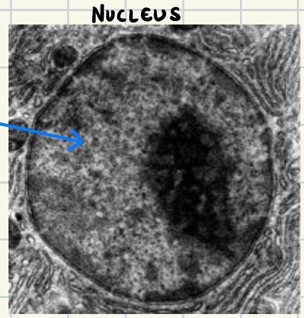
structure of a chromosome :-

- DNA
- histone proteins
- sister chromatids
- centromere
- telomeres

DNA bound to histones

chromatin

NUCLEUS



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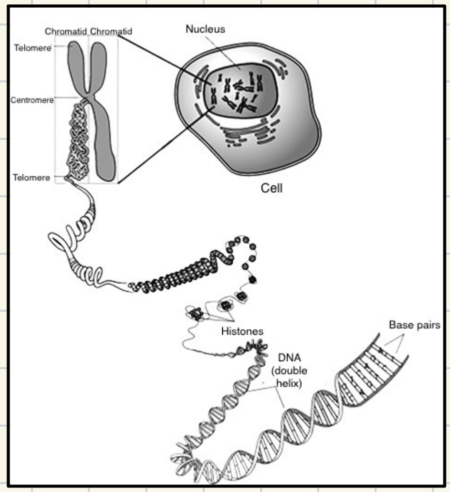


→ 1 chromosome has several thousand genes

→ genes = DNA that codes for protein

→ locus = position of gene in a chromosome

→ the same species have genes on same chromosome & loci (plural of locus)

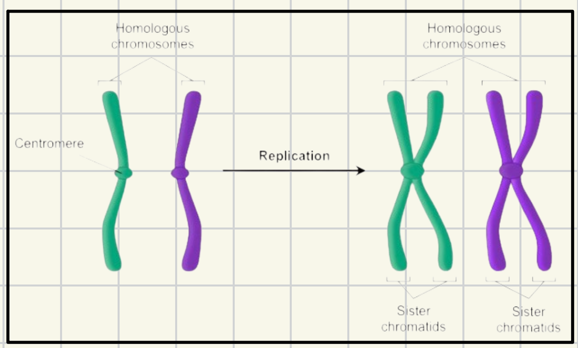
→ alleles = different forms of one gene [due to mutation] & they can be dominant or recessive

homologous chromosomes :-

→ pairs of chromosomes found in diploid cells

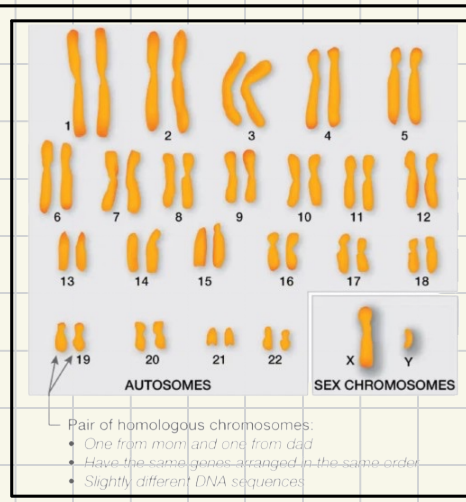
→ one maternal & one paternal

→ similar centrosome position, chromosome size/shape and same genes but may have different alleles



Karyogram →

- total set of chromosomes in an organism
- arranged in order of size
- # of chromosomes are specific to each species
- human diploid cells have 23 sets of chromosomes
  - ↳ total 46 chromosomes



cell division :

→ all cells in our body are genetically identical except gametes
→ cells reproduce/divide to pass copies of genes to daughter cells

• haploid cells are a result of meiosis and they have only one set of chromosomes and in humans that is 23 sets of chromosomes (gametes / germ cells)
• diploid cells are a result of mitosis, one maternal & one paternal and all cells are diploid except gametes/germ cells and humans that is 46 total chromosomes

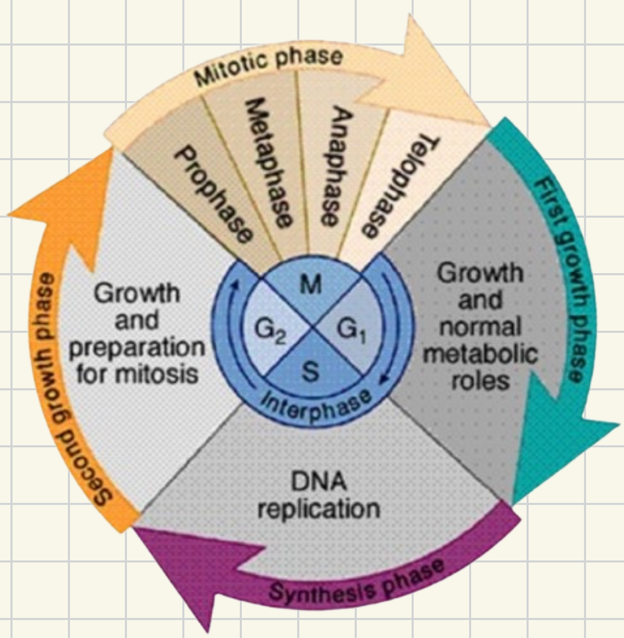
importance of mitosis :

* mitosis results in two genetically identical daughter cells that have the same number of chromosomes as parent cell

1) maintaining number of chromosomes
 └ ensuring genetic stability so new cells retain function
2) growth of multicellular organisms
 └ organisms start from one cell and that cell multiplies by mitosis for growth
3) cell replacement & tissue repair
 └ if a cell dies, another cell can multiply by mitosis to replace it
4) asexual reproduction
 └ [esp. for plants] plants with good traits can have offspring with the same traits

the mitotic cell cycle :

1) Interphase
 └ G_1 → growth
 └ S → synthesis (cell division)
 └ G_2 → growth
2) M phase (mitosis)
 └ prophase
 └ metaphase
 └ anaphase
 └ telophase
3) cytokinesis



* interphase

1) G_1 phase

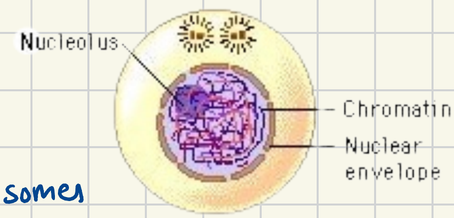
L growth of cell & synthesis of proteins and other substances

2) S phase

L DNA replication which results in sister chromatids

3) G_2 phase

L growth & duplication of centrioles forming 2 pairs of centrosomes & repair of DNA replication errors in the S phase



* M phase (mitosis)

1) prophase

L condensation of chromatin; it becomes shorter & thicker, making the sister chromatids & chromosome visible

L nucleolus disappears & nuclear membrane starts to break down

L a pair of centrioles move to either pole of the cell

L the pairs of centrioles begin to make spindle fibres

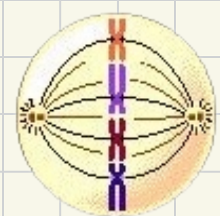
* plant cells don't have centrioles but other structures in the cell help form the spindle fibres

2) metaphase

L centrosomes reach poles & spindle fibres fully form

L the spindle fibres are released from centrioles and attach to the centromere of each chromosome

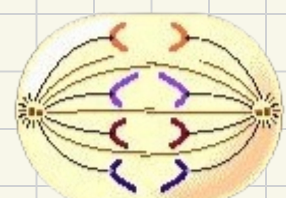
L the spindle fibres allign the chromosomes along the equator/metaphase plate of the cell



3) anaphase * requires energy in the form of ATP

L spindle fibres start to shorten and move towards the centrioles, pulling the centromere & chromatids that they are bound to, towards opposite poles

L this causes the centromere to divide into two & the individual chromatids are pulled to each opposite pole



4) telophase

L the chromosomes are now at each pole of the cell & become longer and thinner again

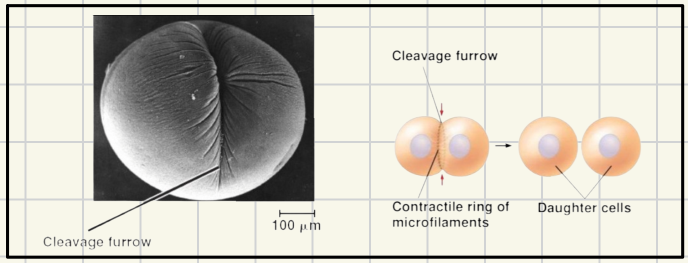
L spindle fibres disintegrate & nuclear membrane reforms

* cytokinesis

→ division of the cell's cytoplasm

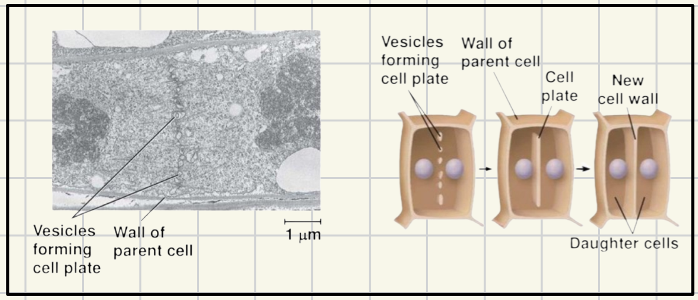
⇒ in animal cells

- the contractile ring of microfilaments form a cleavage furrow & the cytoskeleton causes the cell membrane to draw inwards until the cell is split into two



⇒ in plant cells

- vesicles, which are cellular envelopes that transport materials, are transported to the equator to form a cell plate which later forms a cell wall, thus splitting the cytoplasm in two and the organelles are shared out

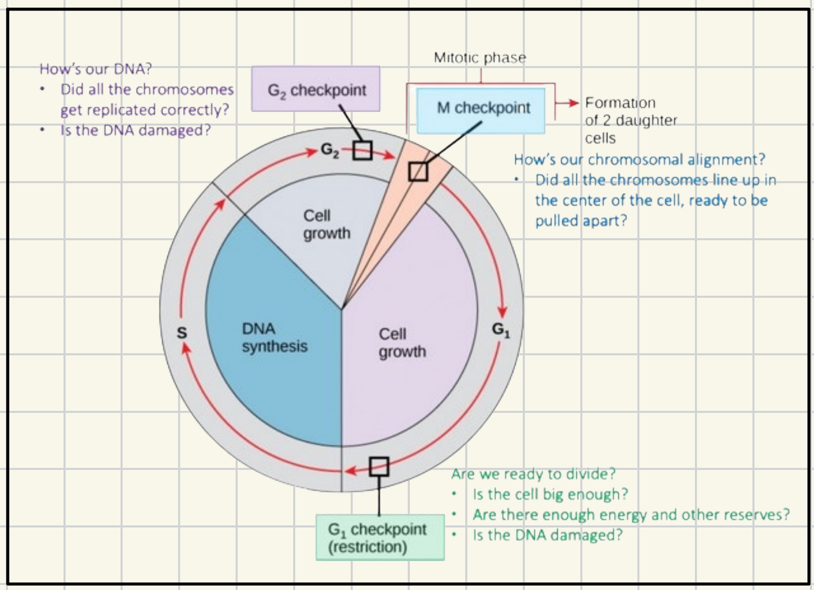


Checkpoints :

- the duration of the cell cycle differs for different cells

- the cell cycle is tightly controlled and coordinated

- checkpoints prevent the cycle from continuing when a mistake is made or DNA is damaged



telomere :

→ a region of repetitive nucleotide sequences at the ends of a chromatid

→ they don't code for protein so they have no genes

Role : DNA replicating enzymes stop a little before the end of DNA molecules so having the telomeres at the end prevents loss of genes at the tips of each chromatid

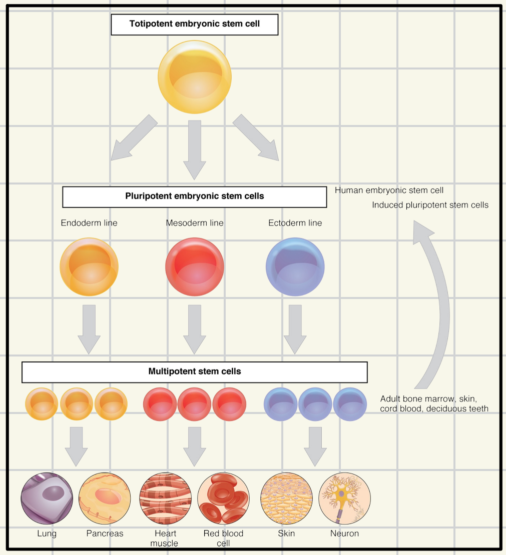
BUT with each mitotic division, the telomeres get shorter & shorter so after many rounds of mitosis, aged cells eventually lose vital genes and die

(!) when cells age, we age too. until they die, then we die too

can we repair telomeres?	the enzyme telomerase repairs telomeres, which stops it from getting shorter each time mitosis takes place BUT telomerase is not active in human body cells however it is active in stem cells & cancer cells

Stem cells :: no specific function

TYPES: Totipotent, Pluripotent, Multipotent



→ unspecialised cells that are able to continually divide by mitosis because they have telomerase, so their telomeres do not shorten each cell cycle

→ when it divides, either another stem cell or a specialised cell is produced

ROLE: forms specialised cells & divides to give continuous supply of stem cells for cell replacement, tissue repair & growth

→ Totipotent
- can divide & produce any type of body cell
- in development, the cells translate only part of the DNA, resulting in cell specialization
- occur only for a limited time in early mammalian embryo

→ Pluripotent
- found in embryos & can become almost any type of cell
- used in research for use to treat human disorders but treatment can fail or stem cells can continually divide to create tumors

→ Multipotent
- found in mature mammals & can divide to form a limited number of different cell types

potential uses of stem cells	stem cells can be used in both research & medicine, for repairing dead tissues or treatment of neurological conditions such as Alzheimer's, as well as research in developmental biology. e.g removing some stem cells from a healthy donor and injecting it in the bone marrow of another person that is suffering from an injury or disease

Cancer cells ::

→ a result of uncontrolled rapid mitosis because when cells divide repeatedly, the cell cycle checkpoints are not controlled

→ due to this, there is a short interphase thus there is little growth & DNA replication is error-prone

<u>Factors that ↑ risk of cancer</u>

- mutagens = substances that cause mutation
- carcinogens = substances that cause cancer

i.e ionising radiation (X-rays etc.), UV light, free radicals, chemicals (tar, mustard gas etc.), virus infection (HIV, AIDS)

- hereditary predisposition, tobacco smoking & obesity are also risk-increasing factors

* all carcinogens are mutagens but all mutagens are not carcinogens

* cancer is not usually due to a single mutation in a cell

Cause of cancer :

→ a mutation in genes that controls cell division

→ proto-oncogenes are normal genes that become oncogenes (cancer-causing) by mutation

→ tumor suppresor genes are switched off as a result

→ thus, cancer cells pass on mutations & oncogenes to daughter cells as they can escape DNA repair during interphase as well

→ this results in accumulated mutations

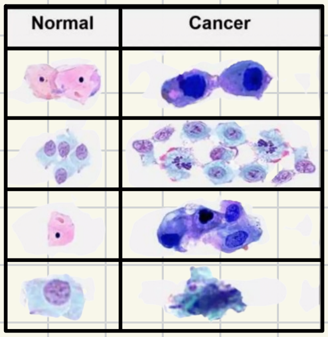
- immune system cannot recognize the cells as foreign, thus cannot destroy them
- no programmed cell death (cell cannot die)
- telomerase can get activated, preserving the telomeres, allowing cancer cells to continually divide
- do not respond to intra/extracellular signals to stop cell division
- prevents contact inhibition, which is the prevention of mitosis by cell to cell contact
- loss of cell specialization / function

formation of mass of unspecialized cells = $\boxed{\text{TUMOUR}}$

⇒ as the tumour grows, it displaces & compresses functional surrounding tissues

⇒ blood capillaries grow into tumour as it needs a lot of nutrients to grow

angiogenesis

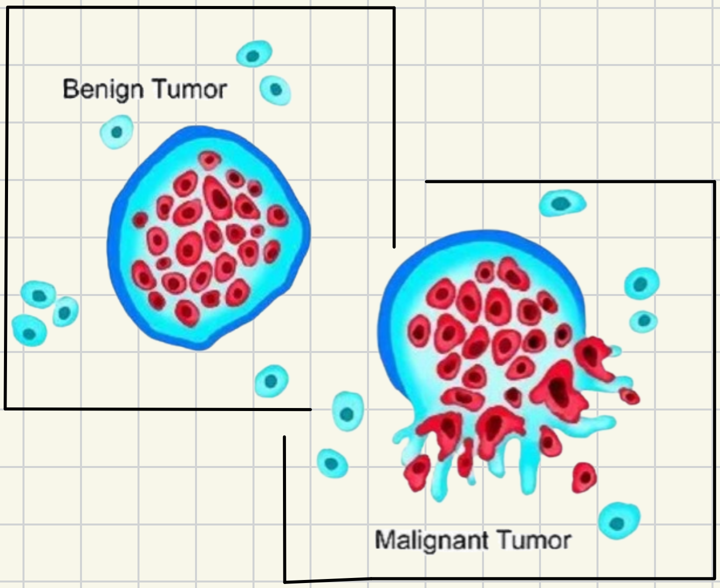


→ abnormal change in cell shape

→ no specific functions of cells

→ there are 2 types of tumours

Types of tumours



✱ Benign

└ doesn't spread from site of origin

e.g warts, ovarian cysts, brain tumors

✱ Malignant

└ can spread throughout the body (metastasis)

└ spread via blood & lymphatic system

└ invades & destroys other tissues and can result in secondary growth